

1. Questions

Read the following data carefully and answer the questions based on it-

Some number of students appeared for three exams (Bank, SSC, and CAT) from five different cities (A, B, C, D, and E). The table given below shows the average number of students who appeared for the bank and SSC exams and the average number of students who appeared for the SSC and CAT exams from each city.

City	The average number of students who appeared for the bank and SSC exam	The average number of students who appeared for the SSC and CAT exam
A	2950	2865
B	2425	2250
C	2975	2875
D	2900	2990
E	2725	2430

Notes:

The ratio of the number of students who appeared for the bank and CAT exams from city A is 20:19, respectively. The average number of students who appeared for all three exams from cities B and C together is 8150, which is $(Y^3 - 20)$ less than the total number of students who appeared from city A. The average number of students who appeared for the bank exam from cities B and E together is $(Y^2 + 10Y)$ more than the average number of students who appeared for the SSC exam from cities D and E together. The ratio of the number of students who appeared for the CAT exam from cities D and E is 48:41. The average number of students who appeared for the SSC exam from cities B and E is 2200.

The total number of students who appeared for the bank exam from cities A and D together is approximately what percent of the average number of students who appeared for all three exams from cities B and C?

- 69%
- 75%
- 65%
- 72%
- None of these

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2. Questions

Read the following data carefully and answer the questions based on it-

Some number of students appeared for three exams (Bank, SSC, and CAT) from five different cities (A, B, C, D, and E). The table given below shows the average number of students who appeared for the bank and

SSC exams and the average number of students who appeared for the SSC and CAT exams from each city.

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The difference between the number of students who appeared for the bank and CAT exams from city B is “P+90” and the difference between the number of students who appeared for the bank and CAT exams from city D is “Q – 60”. Find the value of $0.5Y(P-Q)$.

- 90
- 120
- 110
- 100
- None of these

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3. Questions

Read the following data carefully and answer the questions based on it-

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If “N%” of students who appeared for the SSC exam from city C and “(N +5)%” of students who appeared for the SSC exam from city E are females and the total number of males who appeared for the SSC exam from cities C and E together is 2970, then find the number of males who appeared for the SSC exam from city E.

- 1320
- 1260
- 1440
- 1280
- 1560

4. Questions

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Read the following data carefully and answer the questions based on it-

Some number of students appeared for three exams (Bank, SSC, and CAT) from five different cities (A, B, C, D, and E). The table given below shows the average number of students who appeared for the bank and SSC exams and the average number of students who appeared for the SSC and CAT exams from each city.

City	The average number of students who appeared for the bank and SSC exam	The average number of students who appeared for the SSC and CAT exam
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Notes:

The ratio of the number of students who appeared for the bank and CAT exams from city A is 20:19, respectively. The average number of students who appeared for all three exams from cities B and C together is 8150, which is $(Y^3 - 20)$ less than the total number of students who appeared from city A. The average number of students who appeared for the bank exam from cities B and E together is $(Y^2 + 10Y)$ more than the average number of students who appeared for the SSC exam from cities D and E together. The ratio of the number of students who appeared for the CAT exam from cities D and E is 48:41. The average number of students who appeared for the SSC exam from cities B and E is 2200.

The number of students who appeared for all three exams from city F is “Y%” more than that of city D. 25% of the total number of students appeared for exams from city F, appeared for bank exams, and the number of students who appeared for the CAT exam from city F and city B is the same. Find the difference between the number of students who appeared for the SSC and bank exams from city F.

- 2268
- 2312
- 2274
- 2194
- 2294

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5. Questions

Read the following data carefully and answer the questions based on it-

Some number of students appeared for three exams (Bank, SSC, and CAT) from five different cities (A, B, C, D, and E). The table given below shows the average number of students who appeared for the bank and SSC exams and the average number of students who appeared for the SSC and CAT exams from each city.

City	The average number of students who appeared for the bank and SSC exam	The average number of students who appeared for the SSC and CAT exam
A	2950	2865
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D	2900	2990
E	2725	2430

Notes:

The ratio of the number of students who appeared for the bank and CAT exams from city A is 20:19, respectively. The average number of students who appeared for all three exams from cities B and C together is 8150, which is $(Y^3 - 20)$ less than the total number of students who appeared from city A. The average number of students who appeared for the bank exam from cities B and E together is $(Y^2 + 10Y)$ more than the average number of students who appeared for the SSC exam from cities D and E together. The ratio of the number of students who appeared for the CAT exam from cities D and E is 48:41. The average number of students who appeared for the SSC exam from cities B and E is 2200.

Find the difference between the average number of students who appeared for the bank and SSC exams from all five cities together.

- 460
- 490
- 510
- 530
- None of these

6. Questions

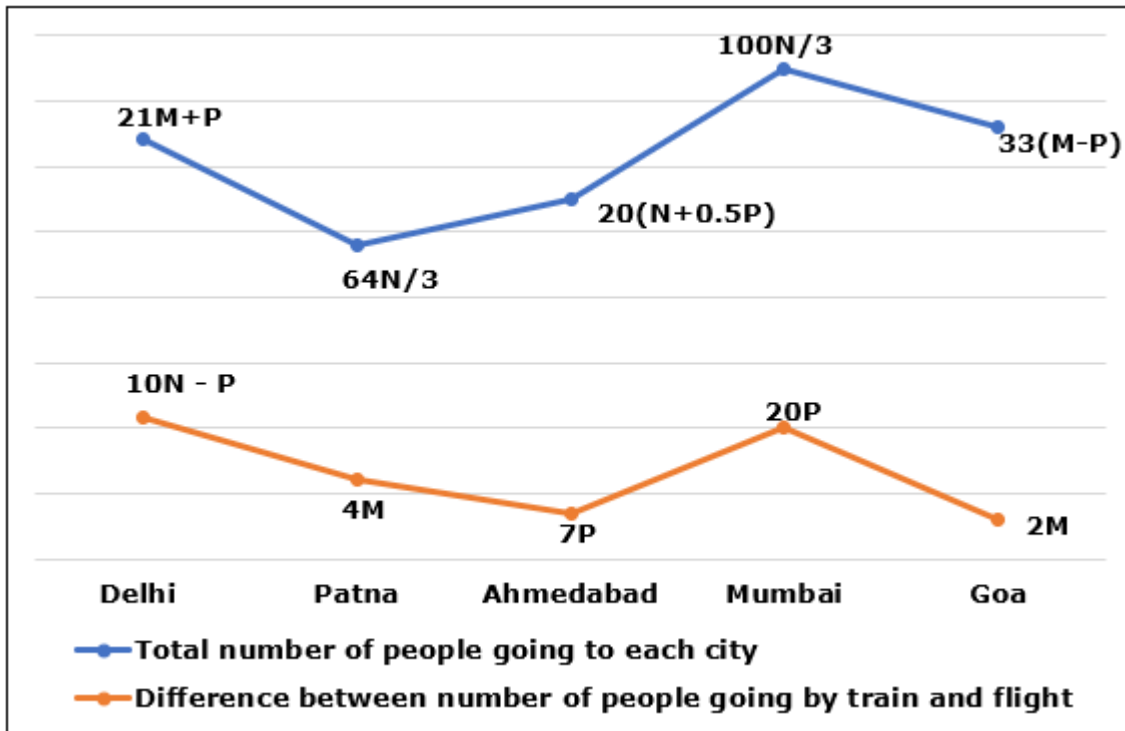
Out of the total number of students who appeared for all three exams from city B, 20% of students attended for all three exams, which is 1470 less than the number of students who attended for only one exam. Then find the percentage of students who attended for exactly two exams from city B.

- 30%
- 35%
- 45%
- 40%
- None of these

7. Questions

Read the following data carefully and answer the questions based on it-

There are some number of people who are going to five different cities, i.e., Delhi, Patna, Ahmedabad, Mumbai, and Goa. They are either going by train or flight. The line graph given below shows the number of people going to each city and the difference between the number of people going by train and flight. The number of people going by flight is less than that of by train.



Note:

The average number of people going to Delhi and Goa together is 1300. The ratio of the number of people going to Mumbai and Ahmedabad by flight is 55:48, and the number of people going to Delhi by train is 135 more than that of Goa.

Find the correct combination between column I and column II.

Column I	Column II
A). Ratio of the number of people going to Patna and Goa by train.	P). 3:4
B). Ratio of total number of people going to Mumbai and Patna.	Q). 5:6
C). Ratio of the number of people going to Patna and Ahmedabad by flight.	R). 25:16

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a. A-Q, B-P and C-R

b. A-P, B-R and C-Q

- c. A-Q, B-R and C-P
- d. A-R, B-P and C-Q
- e. None of these

8. Questions

If the difference between the total number of people going to Mumbai and Ahmedabad is 40% of the total number of people going to Kolkata and the average number of people going to Goa and Kolkata by flight is 475, then find the number of people going to Kolkata by train.

- a. 650
- b. 575
- c. 720
- d. 550
- e. None of these

9. Questions

If the average number of people going to Delhi, Goa, and Patna by train is $(X + 250)$ and the average number of people going to Ahmedabad and Mumbai by flight is $(Y - 105)$, then which among the following options is true about the value of $(Y - X)$?

I: Sum of the digits of $(Y - X)$ is an even number.

II: At least one of the digits of $(Y - X)$ is a perfect square.

III: It has 6 factors.

- a. Only I
- b. Only II
- c. Only III
- d. Both I and II
- e. All I, II and III

10. Questions

What is the sum of $(P)\%$ of the total number of people going to all five cities by flight and $[(N+5)/2]\%$ of the average number of people going to all five cities together?

- a. 625
- b. 881
- c. 436
- d. 791

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e. 888

11. Questions

Out of the total number of people going to Delhi by train, 40% are females, and the total number of females going to Delhi is 542. Find the difference between the number of males going to Delhi by train and flight.

- a. $4.6M + 22$
- b. $5M - 12$
- c. $6N + 28$
- d. Both (b) and (c)
- e. Both (a) and (b)

12. Questions

The average number of people going to Patna and Delhi is approximately what percent less or more than the average number of people going to Mumbai and Goa?

- a. 26%
- b. 18%
- c. 21%
- d. 31%
- e. None of these

13. Questions

In each question two equations are given below. Solve them and answer the questions.

(I): $x^3 + Ax^2 + 19x + 12 = 0$, where A is a cube of smallest prime number.

(II): $4y^3 - (B+12)y^2 + 68y - 2B = 0$, where B is 150% more than that of A.

Establish the relationship between the values of x and y.

- a. $x > y$
- b. $x < y$
- c. $x \leq y$
- d. $x \geq y$
- e. $x = y$ or can't be determined

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14. Questions

(I): $3x^3 - 5x^2 - 11x = 3$

(II): $2y^3 + y^2 - 5y = -2$

Establish the relationship between value of x and y.

- a. $x > y$
- b. $x < y$
- c. $x \leq y$
- d. $x \geq y$
- e. $x = y$ or can't be determined

15. Questions

(I): $x^3 + Qx^2 - (P-Q)x = 0$, where sum of the roots of equation is -5 and value of P is 4 more than thrice the value of Q.

(II): $y^3 - Ry^2 + (R-1)y = 0$, where product of roots is 3

Establish the relationship between value of x and y.

- a. $x > y$
- b. $x < y$
- c. $x \leq y$
- d. $x \geq y$
- e. $x = y$ or can't be determined

16. Questions

Find the which of the following statements is true?

I: $X^2 + (P + 2)A - 12P = 0$

Note: The negative root of the equation is -12.

I: P is the prime number.

II: The smallest root of the equation is a prime number.

III: 'P + 4' is a perfect square.

- a. Only I
- b. All of them
- c. Both I and II
- d. Only II
- e. Both I and III

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17. Questions

(I): $x^2 - 29x + 52A = 0$

(II): $y^2 - 38y + 65B = 0$

Note: One of the roots in equation I is 16, and the smallest root of both equations is the same. Find the sum of the positive roots of both the equations.

- a. 41
- b. 29
- c. 38
- d. 34
- e. None of these

18. Questions

Read the following data carefully and answer the questions based on it-

Train A can cross a bridge of length “X” metre in 9 seconds. Train B running with the speed of (Y-20) km/h can cross a pole in 5.76 seconds and cross a platform of length (X+24) metre in 14.4 seconds. Train C left station P for station Q at 8 am, and in its whole journey to reach station Q, train C took three halts of equal duration. If train C did not stop anywhere between station P and station Q, then it could reach its destination 1 hour before. The length of train A is 150 metre which is $(X/2 - 6)$ metre more than that of train B. Train B and train C, both running in the same direction, can cross each other in 82.8 seconds, and speed of train C is 20m/s.

If train C reaches station Q at 5:20 p.m., then find the difference between the time taken by train A and train B to travel the distance between station P and station Q.

- a. $3(4/9)$ hours
- b. $4(2/9)$ hours
- c. $5(1/3)$ hours
- d. $4(4/9)$ hours
- e. $6(1/3)$ hours

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19. Questions

If a person is running with the speed of 6 m/s in the same direction as train B, then find the time taken by train B to completely cross the person.

- a. 12 seconds
- b. 9 seconds
- c. 15 seconds

- d. 7 seconds
- e. None of these

20. Questions

Train C can cross the bridge in “M” seconds, and the time taken by train A to cross the platform is (N + 4.8) seconds. Find the time taken by train C to cross a tunnel whose length is equal to that of train A in terms of “M” and “N.”

- a. $(2N + 5.5)$ s
- b. $(2M - 3N)$ s
- c. $(M/10 + 3N)$ s
- d. Both B and C
- e. None of these

Explanations:

1. Questions

Number of students who appeared for the bank and SSC exam from city A = $2950 \times 2 = 5900$

Number of students who appeared for the SSC and CAT exam from city A = $2865 \times 2 = 5730$

Ratio of number of students who appeared for the bank and CAT exam from city A is 20:19

Let number of students who appeared for the SSC exam from city A = X

$$(5900 - X)/(5730 - X) = 20/19$$

$$112100 - 19X = 114600 - 20X$$

$$X = 2500$$

Number of students who appeared for the SSC exam from city A = 2500

Number of students who appeared for the bank exam from city A = $5900 - 2500 = 3400$

Number of students who appeared for the CAT exam from city A = $5730 - 2500 = 3230$

Total number of students who appeared for all three exams from city A = $3400 + 2500 + 3230 = 9130$

Total number of students who appeared for all three exams from city B and C = $8150 \times 2 = 16300$

$$9130 - 8150 = 980$$

$$Y^3 - 20 = 980$$

$$Y^3 = 1000$$

$$Y = 10$$

The number of students who appeared for the bank and SSC exam from cities B and E together = $4850 + 5450 = 10300$

Let the number of students who appeared for the bank exam from cities B and E together = P

The number of students who appeared for the SSC exam from cities B and E together = $10300 - P$

The average number of students who appeared for the SSC exam from cities B and E is 2200:

$$10300 - P = 2200 * 2$$

$$10300 - P = 4400$$

$$P = 5900$$

Total number of students who appeared for the bank exam from cities B and E together = 5900

Average number of students who appeared for the bank exam from cities B and E together = $5900 / 2 = 2950$

Now,

$$Y^2 + 10Y = 10^2 + 10(10) = 100 + 100 = 200$$

Average number of students who appeared for the SSC exam from cities D and E together = $2950 - 200 = 2750$

Total number of students who appeared for the SSC exam from cities D and E together = $2750 * 2 = 5500$

Total number of students who appeared for the bank and SSC exam from cities D and E together = $2 * 2990 + 2 * 2430 = 5980 + 4860 = 10840$

Total number of students who appeared for the bank exam from cities D and E together = $10840 - 5500 = 5340$

Total number of students who appeared for the CAT exam from cities D and E together = $(2 * 2990 + 2 * 2430) - 5500 = 5980 + 4860 - 5500 = 5340$

Number of students who appeared for the CAT exam from city D = $5340 / 89 * 48 = 2880$

Number of students who appeared for the CAT exam from city E = $5340 / 89 * 41 = 2460$

Number of students who appeared for the SSC exam from city D = $5980 - 2880 = 3100$

Number of students who appeared for the bank exam from city D = $5800 - 3100 = 2700$

Number of students who appeared for the SSC exam from city E = $4860 - 2460 = 2400$

Number of students who appeared for the bank exam from city E = $5450 - 2400 = 3050$

Number of students who appeared for the SSC exam from city B = $4400 - 2400 = 2000$

Number of students who appeared for the bank exam from city B = $4850 - 2000 = 2850$

Number of students who appeared for the CAT exam from city B = $4500 - 2000 = 2500$

Total number of students who appeared for all three exams from city B = $2850 + 2000 + 2500 = 7350$

Total number of students who appeared for all three exams from city C = $16300 - 7350 = 8950$

Number of students who appeared for the CAT exam from city C = $8950 - 5950 = 3000$

Number of students who appeared for the SSC exam from city C = $5750 - 3000 = 2750$

Number of students who appeared for the bank exam from city C = $5950 - 2750 = 3200$

City	Number of students who appeared for the bank exam	Number of students who appeared for the SSC exam	Number of students who appeared for the CAT exams	Total number of students
A	3400	2500	3230	9130
B	2850	2000	2500	7350
C	3200	2750	3000	8950
D	2700	3100	2880	8680
E	3050	2400	2460	7910

Answer: B

Total number of students who appeared for the bank exam from cities A and D together = $3400 + 2700 = 6100$

Average number of students who appeared for all three exams from cities B and C = 8150

Required percentage = $6100/8150 \times 100$

= 74.84% ~ 75%

Hence the answer is option B.

2. Questions

Number of students who appeared for the bank and SSC exam from city A = $2950 \times 2 = 5900$

Number of students who appeared for the SSC and CAT exam from city A = $2865 \times 2 = 5730$

Ratio of number of students who appeared for the bank and CAT exam from city A is 20:19

Let number of students who appeared for the SSC exam from city A = X

$(5900 - X)/(5730 - X) = 20/19$

$112100 - 19X = 114600 - 20X$

$X = 2500$

Number of students who appeared for the SSC exam from city A = 2500

Number of students who appeared for the bank exam from city A = $5900 - 2500 = 3400$

Number of students who appeared for the CAT exam from city A = $5730 - 2500 = 3230$

Total number of students who appeared for all three exams from city A = $3400 + 2500 + 3230 = 9130$

Total number of students who appeared for all three exams from city B and C = $8150 \times 2 = 16300$

$9130 - 8150 = 980$

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$$Y^3 - 20 = 980$$

$$Y^3 = 1000$$

$$Y = 10$$

The number of students who appeared for the bank and SSC exam from cities B and E together = $4850 + 5450 = 10300$

Let the number of students who appeared for the bank exam from cities B and E together = P

The number of students who appeared for the SSC exam from cities B and E together = $10300 - P$

The average number of students who appeared for the SSC exam from cities B and E is 2200:

$$10300 - P = 2200 * 2$$

$$10300 - P = 4400$$

$$P = 5900$$

Total number of students who appeared for the bank exam from cities B and E together = 5900

Average number of students who appeared for the bank exam from cities B and E together = $5900 / 2 = 2950$

Now,

$$Y^2 + 10Y = 10^2 + 10(10) = 100 + 100 = 200$$

Average number of students who appeared for the SSC exam from cities D and E together = $2950 - 200 = 2750$

Total number of students who appeared for the SSC exam from cities D and E together = $2750 * 2 = 5500$

Total number of students who appeared for the bank and SSC exam from cities D and E together = $2 * 2990 + 2 * 2430 = 5980 + 4860 = 10840$

Total number of students who appeared for the bank exam from cities D and E together = $10840 - 5500 = 5340$

Total number of students who appeared for the CAT exam from cities D and E together = $(2 * 2990 + 2 * 2430) - 5500 = 5980 + 4860 - 5500 = 5340$

Number of students who appeared for the CAT exam from city D = $5340 / 89 * 48 = 2880$

Number of students who appeared for the CAT exam from city E = $5340 / 89 * 41 = 2460$

Number of students who appeared for the SSC exam from city D = $5980 - 2880 = 3100$

Number of students who appeared for the bank exam from city D = $5800 - 3100 = 2700$

Number of students who appeared for the SSC exam from city E = $4860 - 2460 = 2400$

Number of students who appeared for the bank exam from city E = $5450 - 2400 = 3050$

Number of students who appeared for the SSC exam from city B = $4400 - 2400 = 2000$

Number of students who appeared for the bank exam from city B = $4850 - 2000 = 2850$

Number of students who appeared for the CAT exam from city B = $4500 - 2000 = 2500$

Total number of students who appeared for all three exams from city B = $2850 + 2000 + 2500 = 7350$

Total number of students who appeared for all three exams from city C = $16300 - 7350 = 8950$

Number of students who appeared for the CAT exam from city C = $8950 - 5950 = 3000$

Number of students who appeared for the SSC exam from city C = $5750 - 3000 = 2750$

Number of students who appeared for the bank exam from city C = $5950 - 2750 = 3200$

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C	3200	2750	3000	8950
D	2700	3100	2880	8680
E	3050	2400	2460	7910

Answer: D

The difference between number of students who appeared for the bank and CAT exam from city B = $2850 - 2500 = 350$

$P + 90 = 350$

$P = 260$

The difference between number of students who appeared for the bank and CAT exam from city D = $2880 - 2700 = 180$

$Q - 60 = 180$

$Q = 240$

Let

$0.5Y(P - Q)$

$0.5 \times 10(260 - 240)$

$= 100$

Hence the answer is option D.

3. Questions

Number of students who appeared for the bank and SSC exam from city A = $2950 \times 2 = 5900$

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Number of students who appeared for the SSC and CAT exam from city A = $2865 \times 2 = 5730$

Ratio of number of students who appeared for the bank and CAT exam from city A is 20:19

Let number of students who appeared for the SSC exam from city A = X

$$(5900 - X)/(5730 - X) = 20/19$$

$$112100 - 19X = 114600 - 20X$$

$$X = 2500$$

Number of students who appeared for the SSC exam from city A = 2500

Number of students who appeared for the bank exam from city A = $5900 - 2500 = 3400$

Number of students who appeared for the CAT exam from city A = $5730 - 2500 = 3230$

Total number of students who appeared for all three exams from city A = $3400 + 2500 + 3230 = 9130$

Total number of students who appeared for all three exams from city B and C = $8150 \times 2 = 16300$

$$9130 - 8150 = 980$$

$$Y^3 - 20 = 980$$

$$Y^3 = 1000$$

$$Y = 10$$

The number of students who appeared for the bank and SSC exam from cities B and E together = $4850 + 5450 = 10300$

Let the number of students who appeared for the bank exam from cities B and E together = P

The number of students who appeared for the SSC exam from cities B and E together = $10300 - P$

The average number of students who appeared for the SSC exam from cities B and E is 2200:

$$10300 - P = 2200 \times 2$$

$$10300 - P = 4400$$

$$P = 5900$$

Total number of students who appeared for the bank exam from cities B and E together = 5900

Average number of students who appeared for the bank exam from cities B and E together = $5900 / 2 = 2950$

Now,

$$Y^2 + 10Y = 10^2 + 10(10) = 100 + 100 = 200$$

Average number of students who appeared for the SSC exam from cities D and E together = $2950 - 200 = 2750$

Total number of students who appeared for the SSC exam from cities D and E together = $2750 \times 2 = 5500$

Total number of students who appeared for the bank and SSC exam from cities D and E together = $2 * 2900 + 2 * 2725 = 5800 + 5450 = 11250$

$$2900 + 2 * 2725 = 5800 + 5450 = 11250$$

Total number of students who appeared for the bank exam from cities D and E together = $11250 - 5500 = 5750$

Total number of students who appeared for the CAT exam from cities D and E together = $(2 * 2990 + 2 * 2430) - 5500 = 5980 + 4860 - 5500 = 5340$

Number of students who appeared for the CAT exam from city D = $5340/89*48 = 2880$

Number of students who appeared for the CAT exam from city E = $5340/89*41 = 2460$

Number of students who appeared for the SSC exam from city D = $5980 - 2880 = 3100$

Number of students who appeared for the bank exam from city D = $5800 - 3100 = 2700$

Number of students who appeared for the SSC exam from city E = $4860 - 2460 = 2400$

Number of students who appeared for the bank exam from city E = $5450 - 2400 = 3050$

Number of students who appeared for the SSC exam from city B = $4400 - 2400 = 2000$

Number of students who appeared for the bank exam from city B = $4850 - 2000 = 2850$

Number of students who appeared for the CAT exam from city B = $4500 - 2000 = 2500$

Total number of students who appeared for all three exams from city B = $2850 + 2000 + 2500 = 7350$

Total number of students who appeared for all three exams from city C = $16300 - 7350 = 8950$

Number of students who appeared for the CAT exam from city C = $8950 - 5950 = 3000$

Number of students who appeared for the SSC exam from city C = $5750 - 3000 = 2750$

Number of students who appeared for the bank exam from city C = $5950 - 2750 = 3200$

City	Number of students who appeared for the bank exam	Number of students who appeared for the SSC exam	Number of students who appeared for the CAT exams	Total number of students
A	3400	2500	3230	9130
B	2850	2000	2500	7350
C	3200	2750	3000	8950
D	2700	3100	2880	8680
E	3050	2400	2460	7910

Answer: A

Total number of males who appeared for the SSC exam from city C and E together = 2970

Total number of females who appeared for the SSC exam from city C and E together = $5150 - 2970 = 2180$

$$2750 * N\% + 2400 * (N+5)\% = 2180$$

$$275N + 240N + 1200 = 21800$$

$$515N = 20600$$

$$N = 40$$

$$\begin{aligned}\text{Number of males who appeared for the SSC exam from city E} &= 2400 \times 55\% \\ &= 1320\end{aligned}$$

Hence the answer is option A.

4. Questions

$$\text{Number of students who appeared for the bank and SSC exam from city A} = 2950 \times 2 = 5900$$

$$\text{Number of students who appeared for the SSC and CAT exam from city A} = 2865 \times 2 = 5730$$

$$\text{Ratio of number of students who appeared for the bank and CAT exam from city A is } 20:19$$

$$\text{Let number of students who appeared for the SSC exam from city A} = X$$

$$(5900 - X)/(5730 - X) = 20/19$$

$$112100 - 19X = 114600 - 20X$$

$$X = 2500$$

$$\text{Number of students who appeared for the SSC exam from city A} = 2500$$

$$\text{Number of students who appeared for the bank exam from city A} = 5900 - 2500 = 3400$$

$$\text{Number of students who appeared for the CAT exam from city A} = 5730 - 2500 = 3230$$

$$\text{Total number of students who appeared for all three exams from city A} = 3400 + 2500 + 3230 = 9130$$

$$\text{Total number of students who appeared for all three exams from city B and C} = 8150 \times 2 = 16300$$

$$9130 - 8150 = 980$$

$$Y^3 - 20 = 980$$

$$Y^3 = 1000$$

$$Y = 10$$

$$\begin{aligned}\text{The number of students who appeared for the bank and SSC exam from cities B and E together} &= 4850 + 5450 = 10300\end{aligned}$$

$$\text{Let the number of students who appeared for the bank exam from cities B and E together} = P$$

$$\text{The number of students who appeared for the SSC exam from cities B and E together} = 10300 - P$$

$$\text{The average number of students who appeared for the SSC exam from cities B and E is } 2200:$$

$$10300 - P = 2200 \times 2$$

$$10300 - P = 4400$$

$$P = 5900$$

Total number of students who appeared for the bank exam from cities B and E together = 5900

Average number of students who appeared for the bank exam from cities B and E together = $5900 / 2 = 2950$

Now,

$$Y^2 + 10Y = 10^2 + 10(10) = 100 + 100 = 200$$

Average number of students who appeared for the SSC exam from cities D and E together = $2950 - 200 = 2750$

Total number of students who appeared for the SSC exam from cities D and E together = $2750 * 2 = 5500$

Total number of students who appeared for the bank and SSC exam from cities D and E together = $2 * 2900 + 2 * 2725 = 5800 + 5450 = 11250$

Total number of students who appeared for the bank exam from cities D and E together = $11250 - 5500 = 5750$

Total number of students who appeared for the CAT exam from cities D and E together = $(2 * 2990 + 2 * 2430) - 5500 = 5980 + 4860 - 5500 = 5340$

Number of students who appeared for the CAT exam from city D = $5340/89*48 = 2880$

Number of students who appeared for the CAT exam from city E = $5340/89*41 = 2460$

Number of students who appeared for the SSC exam from city D = $5980 - 2880 = 3100$

Number of students who appeared for the bank exam from city D = $5800 - 3100 = 2700$

Number of students who appeared for the SSC exam from city E = $4860 - 2460 = 2400$

Number of students who appeared for the bank exam from city E = $5450 - 2400 = 3050$

Number of students who appeared for the SSC exam from city B = $4400 - 2400 = 2000$

Number of students who appeared for the bank exam from city B = $4850 - 2000 = 2850$

Number of students who appeared for the CAT exam from city B = $4500 - 2000 = 2500$

Total number of students who appeared for all three exams from city B = $2850 + 2000 + 2500 = 7350$

Total number of students who appeared for all three exams from city C = $16300 - 7350 = 8950$

Number of students who appeared for the CAT exam from city C = $8950 - 5950 = 3000$

Number of students who appeared for the SSC exam from city C = $5750 - 3000 = 2750$

Number of students who appeared for the bank exam from city C = $5950 - 2750 = 3200$

Answer: C

The number of students who appeared for all three exams from city F = $8680*110\% = 9548$

Number of students who appeared for the bank exams = $9548/4 = 2387$

Number of students who appeared for the CAT exam from city F = 2500

Number of students who appeared for the SSC exam from city F = $9548 - 4887$

$$= 4661$$

Required difference = $4661 - 2387$

$$= 2274$$

Hence the answer is option C.

5. Questions

Number of students who appeared for the bank and SSC exam from city A = $2950 \times 2 = 5900$

Number of students who appeared for the SSC and CAT exam from city A = $2865 \times 2 = 5730$

Ratio of number of students who appeared for the bank and CAT exam from city A is 20:19

Let number of students who appeared for the SSC exam from city A = X

$$(5900 - X)/(5730 - X) = 20/19$$

$$112100 - 19X = 114600 - 20X$$

$$X = 2500$$

Number of students who appeared for the SSC exam from city A = 2500

Number of students who appeared for the bank exam from city A = $5900 - 2500 = 3400$

Number of students who appeared for the CAT exam from city A = $5730 - 2500 = 3230$

Total number of students who appeared for all three exams from city A = $3400 + 2500 + 3230 = 9130$

Total number of students who appeared for all three exams from city B and C = $8150 \times 2 = 16300$

$$9130 - 8150 = 980$$

$$Y^3 - 20 = 980$$

$$Y^3 = 1000$$

$$Y = 10$$

The number of students who appeared for the bank and SSC exam from cities B and E together = $4850 + 5450 = 10300$

Let the number of students who appeared for the bank exam from cities B and E together = P

The number of students who appeared for the SSC exam from cities B and E together = $10300 - P$

The average number of students who appeared for the SSC exam from cities B and E is 2200:

$$10300 - P = 2200 \times 2$$

$$10300 - P = 4400$$

$$P = 5900$$

Total number of students who appeared for the bank exam from cities B and E together = 5900

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Average number of students who appeared for the bank exam from cities B and E together = $5900 / 2 = 2950$

Now,

$$Y^2 + 10Y = 10^2 + 10(10) = 100 + 100 = 200$$

Average number of students who appeared for the SSC exam from cities D and E together = $2950 - 200 = 2750$

Total number of students who appeared for the SSC exam from cities D and E together = $2750 * 2 = 5500$

Total number of students who appeared for the bank and SSC exam from cities D and E together = $2 * 2900 + 2 * 2725 = 5800 + 5450 = 11250$

Total number of students who appeared for the bank exam from cities D and E together = $11250 - 5500 = 5750$

Total number of students who appeared for the CAT exam from cities D and E together = $(2 * 2990 + 2 * 2430) - 5500 = 5980 + 4860 - 5500 = 5340$

Number of students who appeared for the CAT exam from city D = $5340/89*48 = 2880$

Number of students who appeared for the CAT exam from city E = $5340/89*41 = 2460$

Number of students who appeared for the SSC exam from city D = $5980 - 2880 = 3100$

Number of students who appeared for the bank exam from city D = $5800 - 3100 = 2700$

Number of students who appeared for the SSC exam from city E = $4860 - 2460 = 2400$

Number of students who appeared for the bank exam from city E = $5450 - 2400 = 3050$

Number of students who appeared for the SSC exam from city B = $4400 - 2400 = 2000$

Number of students who appeared for the bank exam from city B = $4850 - 2000 = 2850$

Number of students who appeared for the CAT exam from city B = $4500 - 2000 = 2500$

Total number of students who appeared for all three exams from city B = $2850 + 2000 + 2500 = 7350$

Total number of students who appeared for all three exams from city C = $16300 - 7350 = 8950$

Number of students who appeared for the CAT exam from city C = $8950 - 5950 = 3000$

Number of students who appeared for the SSC exam from city C = $5750 - 3000 = 2750$

Number of students who appeared for the bank exam from city C = $5950 - 2750 = 3200$

@QuizByMickey

City	Number of students who appeared for the bank exam	Number of students who appeared for the SSC exam	Number of students who appeared for the CAT exams	Total number of students
A	3400	2500	3230	9130
B	2850	2000	2500	7350
C	3200	2750	3000	8950
D	2700	3100	2880	8680
E	3050	2400	2460	7910

Answer: B

Average number of students who appeared for the bank exam from all five cities together = $(3400+2850+3200+2700+3050)/5 = 3040$

Average number of students who appeared for the SSC exam from all five cities together = $(2500+2000+2750+3100+2400)/5 = 2550$

Required difference = $3040 - 2550$
 $= 490$

Hence the answer is option B.

6. Questions

Number of students who appeared for the bank and SSC exam from city A = $2950 \times 2 = 5900$

Number of students who appeared for the SSC and CAT exam from city A = $2865 \times 2 = 5730$

Ratio of number of students who appeared for the bank and CAT exam from city A is 20:19

Let number of students who appeared for the SSC exam from city A = X

$$(5900 - X)/(5730 - X) = 20/19$$

$$112100 - 19X = 114600 - 20X$$

$$X = 2500$$

Number of students who appeared for the SSC exam from city A = 2500

Number of students who appeared for the bank exam from city A = $5900 - 2500 = 3400$

Number of students who appeared for the CAT exam from city A = $5730 - 2500 = 3230$

Total number of students who appeared for all three exams from city A = $3400 + 2500 + 3230 = 9130$

Total number of students who appeared for all three exams from city B and C = $8150 \times 2 = 16300$

$$9130 - 8150 = 980$$

$$Y^3 - 20 = 980$$

$$Y^3 = 1000$$

$$Y = 10$$

The number of students who appeared for the bank and SSC exam from cities B and E together = $4850 + 5450 = 10300$

Let the number of students who appeared for the bank exam from cities B and E together = P

The number of students who appeared for the SSC exam from cities B and E together = $10300 - P$

The average number of students who appeared for the SSC exam from cities B and E is 2200:

$$10300 - P = 2200 * 2$$

$$10300 - P = 4400$$

$$P = 5900$$

Total number of students who appeared for the bank exam from cities B and E together = 5900

Average number of students who appeared for the bank exam from cities B and E together = $5900 / 2 = 2950$

Now,

$$Y^2 + 10Y = 10^2 + 10(10) = 100 + 100 = 200$$

Average number of students who appeared for the SSC exam from cities D and E together = $2950 - 200 = 2750$

Total number of students who appeared for the SSC exam from cities D and E together = $2750 * 2 = 5500$

Total number of students who appeared for the bank and SSC exam from cities D and E together = $2 * 2990 + 2 * 2430 = 5980 + 4860 = 10840$

Total number of students who appeared for the bank exam from cities D and E together = $10840 - 5500 = 5340$

Total number of students who appeared for the CAT exam from cities D and E together = $(2 * 2990 + 2 * 2430) - 5500 = 5980 + 4860 - 5500 = 5340$

Number of students who appeared for the CAT exam from city D = $5340 / 89 * 48 = 2880$

Number of students who appeared for the CAT exam from city E = $5340 / 89 * 41 = 2460$

Number of students who appeared for the SSC exam from city D = $5980 - 2880 = 3100$

Number of students who appeared for the bank exam from city D = $5800 - 3100 = 2700$

Number of students who appeared for the SSC exam from city E = $4860 - 2460 = 2400$

Number of students who appeared for the bank exam from city E = $5450 - 2400 = 3050$

Number of students who appeared for the SSC exam from city B = $4400 - 2400 = 2000$

Number of students who appeared for the bank exam from city B = $4850 - 2000 = 2850$

Number of students who appeared for the CAT exam from city B = $4500 - 2000 = 2500$

Total number of students who appeared for all three exams from city B = $2850 + 2000 + 2500 = 7350$

Total number of students who appeared for all three exams from city C = $16300 - 7350 = 8950$

Number of students who appeared for the CAT exam from city C = $8950 - 5950 = 3000$

Number of students who appeared for the SSC exam from city C = $5750 - 3000 = 2750$

Number of students who appeared for the bank exam from city C = $5950 - 2750 = 3200$

City	Number of students who appeared for the bank exam	Number of students who appeared for the SSC exam	Number of students who appeared for the CAT exams	Total number of students
A	3400	2500	3230	9130
B	2850	2000	2500	7350
C	3200	2750	3000	8950
D	2700	3100	2880	8680
E	3050	2400	2460	7910

Answer: D

Total number students who attended for exams from city B = 7350

Number of students who attended for all three exams = 20% of 7350 = 1470

Number of students who attended for only one exam = $1470 + 1470 = 2940$

Number of students who attended for exactly two exams = $7350 - 1470 - 2940 = 2940$

Required percentage = $2940/7350 \times 100$

=40%

Hence the answer is option D.

7. Questions

Number of people going to Delhi = $21M + P$

Number of people going to Goa = $33(M - P)$

Total number of people going to Delhi and Goa = 2600

$21M + P + 33M - 33P = 2600$

$54M - 32P = 2600$

$27M - 16P = 1300$ ---(I)

Total number of people going to Mumbai = $100N/3$

Number of people going to Mumbai by flight = $(100N/3 - 20P)/2$

Total number of people going to Ahmedabad = $20(N+0.5P)$

Number of people going to Ahmedabad by flight = $(20(N+0.5P) - 7P)/2$

@QuizByMickey

$$\frac{\left[\frac{100N/3 - 20P}{2}\right]}{\left[\frac{20(N + 0.5P) - 7P}{2}\right]} = \frac{55}{48}$$

$$(20N/3 - 4P) \times 48 = 11(20N + 10P - 7P)$$

$$320N - 192P = 220N + 33P$$

$$100N = 225P$$

$$P:N = 4:9$$

$$\text{Let } P = 4a \text{ and } N = 9a$$

$$\text{The number of people going to Delhi by train} = (21M + P + 10N - P)/2$$

$$= 10.5M + 5N$$

$$\text{The number of people going to Goa by train} = (33(M - P) + 2M)/2$$

$$= 16.5M - 16.5P + M = 17.5M - 16.5P$$

$$10.5M + 5N - 17.5M + 16.5P = 135$$

$$45a + 66a - 7M = 135$$

$$111a - 7M = 135 \text{---(II)}$$

$$27M - 16P = 1300 \text{---(I)}$$

$$27M - 64a = 1300 \text{---(I)}$$

From equation (I) and (II)

$$I \times 7 \text{ ---} \rightarrow 189M - 448a = 9100$$

$$II \times 27 \text{ ---} \rightarrow 2997a - 189M = 3645$$

$$2549a = 12745$$

$$a = 5$$

and

$$M = 60$$

WKT

$$P = 4a \text{ and } N = 9a$$

$$N = 45$$

$$P = 20$$

Substitute all the variable values and get into the table.

@QuizByMickey

Cities	The total number of people going to each city	The number of people going by train	The number of people going by train
Delhi	1280	855	425
Patna	960	600	360
Ahmedabad	1100	620	480
Mumbai	1500	950	550
Goa	1320	720	600

Answer: C

A). Ratio of the number of people going to Patna and Goa by train =600:720

=5:6

B). Ratio of total number of people going to Mumbai and Patna =1500:960

= 25:16

C). Ratio of the number of people going to Patna and Ahmedabad by flight =360:480

=3:4

Hence the answer is option C

8. Questions

Number of people going to Delhi = 21M + P

Number of people going to Goa = 33(M – P)

Total number of people going to Delhi and Goa = 2600

$$21M + P + 33M - 33P = 2600$$

$$54M - 32P = 2600$$

$$27M - 16P = 1300 \text{---(I)}$$

Total number of people going to Mumbai = 100N/3

Number of people going to Mumbai by flight = (100N/3 - 20P)/2 @QuizByMickey

Total number of people going to Ahmedabad = 20(N+0.5P)

Number of people going to Ahmedabad by flight = (20(N+0.5P) – 7P)/2

$$\frac{\left[\frac{100N/3 - 20P}{2} \right]}{\left[\frac{20(N+0.5P) - 7P}{2} \right]} = \frac{55}{48}$$

$$(20N/3 - 4P) * 48 = 11(20N + 10P - 7P)$$

$$320N - 192P = 220N + 33P$$

$$100N = 225P$$

$$P:N = 4:9$$

$$\text{Let } P = 4a \text{ and } N = 9a$$

$$\text{The number of people going to Delhi by train} = (21M + P + 10N - P)/2$$

$$= 10.5M + 5N$$

$$\text{The number of people going to Goa by train} = (33(M - P) + 2M)/2$$

$$= 16.5M - 16.5P + M = 17.5M - 16.5P$$

$$10.5M + 5N - 17.5M + 16.5P = 135$$

$$45a + 66a - 7M = 135$$

$$111a - 7M = 135 \text{---(II)}$$

$$27M - 16P = 1300 \text{---(I)}$$

$$27M - 64a = 1300 \text{---(I)}$$

From equation (I) and (II)

$$I * 7 \text{---} \rightarrow 189M - 448a = 9100$$

$$II * 27 \text{---} \rightarrow 2997a - 189M = 3645$$

$$2549a = 12745$$

$$a = 5$$

and

$$M = 60$$

WKT

$$P = 4a \text{ and } N = 9a$$

$$N = 45$$

$$P = 20$$

Substitute all the variable values and get into the table.

@QuizByMickey

Cities	The total number of people going to each city	The number of people going by train	The number of people going by train
Delhi	1280	855	425
Patna	960	600	360
Ahmedabad	1100	620	480
Mumbai	1500	950	550
Goa	1320	720	600

Answer: A

Difference between the total number of people going to Mumbai and Ahmedabad = $1500 - 1100 = 400$

Number of people going to Kolkata = $400/2 \times 5 = 1000$

Total number of people going to Goa and Kolkata together by flight = $475 \times 2 = 950$

Number of people going to Kolkata by flight = $950 - 600 = 350$

Number of people going to Kolkata by train = $1000 - 350$

= 650

Hence the answer is option A.

9. Questions

Number of people going to Delhi = $21M + P$

Number of people going to Goa = $33(M - P)$

Total number of people going to Delhi and Goa = 2600

$$21M + P + 33M - 33P = 2600$$

$$54M - 32P = 2600$$

$$27M - 16P = 1300 \text{---(I)}$$

Total number of people going to Mumbai = $100N/3$

Number of people going to Mumbai by flight = $(100N/3 - 20P)/2$ @QuizByMickey

Total number of people going to Ahmedabad = $20(N + 0.5P)$

Number of people going to Ahmedabad by flight = $(20(N + 0.5P) - 7P)/2$

$$\frac{\left[\frac{100N/3 - 20P}{2} \right]}{\left[\frac{20(N + 0.5P) - 7P}{2} \right]} = \frac{55}{48}$$

$$(20N/3 - 4P) \times 48 = 11(20N + 10P - 7P)$$

$$320N - 192P = 220N + 33P$$

$$100N = 225P$$

$$P:N = 4:9$$

$$\text{Let } P = 4a \text{ and } N = 9a$$

$$\text{The number of people going to Delhi by train} = (21M + P + 10N - P)/2$$

$$= 10.5M + 5N$$

$$\text{The number of people going to Goa by train} = (33(M - P) + 2M)/2$$

$$= 16.5M - 16.5P + M = 17.5M - 16.5P$$

$$10.5M + 5N - 17.5M + 16.5P = 135$$

$$45a + 66a - 7M = 135$$

$$111a - 7M = 135 \text{---(II)}$$

$$27M - 16P = 1300 \text{---(I)}$$

$$27M - 64a = 1300 \text{---(I)}$$

From equation (I) and (II)

$$I * 7 \text{---} \rightarrow 189M - 448a = 9100$$

$$II * 27 \text{---} \rightarrow 2997a - 189M = 3645$$

$$2549a = 12745$$

$$a = 5$$

and

$$M = 60$$

WKT

$$P = 4a \text{ and } N = 9a$$

$$N = 45$$

$$P = 20$$

Substitute all the variable values and get into the table.

@QuizByMickey

Cities	The total number of people going to each city	The number of people going by train	The number of people going by train
Delhi	1280	855	425
Patna	960	600	360
Ahmedabad	1100	620	480
Mumbai	1500	950	550
Goa	1320	720	600

Answer: D

Average number of people going to Delhi, Goa and Patna by train = $(855+600+720)/3 = 725$

$$X + 250 = 725$$

$$X = 475$$

Average number of people going to Ahmedabad and Mumbai by flight = $(480+550)/2 = 515$

$$Y - 105 = 515$$

$$Y = 620$$

Required value = $Y - X$

$$620 - 475 = 145$$

I: Sum of the digits of $(Y-X)$ is an even number = $1+4+5 = 10$ (true)

II: At least one of the digits of $(Y - X)$ is a perfect square = 4 or $1 = 2^2$ or 1^2 (true)

III: It has 6 factors = (1, 5, 29, 145) it has 4 factors only = False

Hence the answer is option D.

10. Questions

Number of people going to Delhi = $21M + P$

Number of people going to Goa = $33(M - P)$

Total number of people going to Delhi and Goa = 2600

$$21M + P + 33M - 33P = 2600$$

$$54M - 32P = 2600$$

$$27M - 16P = 1300 \text{---(I)}$$

Total number of people going to Mumbai = $100N/3$

Number of people going to Mumbai by flight = $(100N/3 - 20P)/2$

Total number of people going to Ahmedabad = $20(N+0.5P)$

@QuizByMickey

Number of people going to Ahmedabad by flight = $(20(N+0.5P) - 7P)/2$

$$\frac{\left[\frac{100N/3 - 20P}{2}\right]}{\left[\frac{20(N+0.5P) - 7P}{2}\right]} = \frac{55}{48}$$

$$(20N/3 - 4P) \cdot 48 = 11(20N + 10P - 7P)$$

$$320N - 192P = 220N + 33P$$

$$100N = 225P$$

$$P:N = 4:9$$

$$\text{Let } P = 4a \text{ and } N = 9a$$

$$\text{The number of people going to Delhi by train} = (21M + P + 10N - P)/2$$

$$= 10.5M + 5N$$

$$\text{The number of people going to Goa by train} = (33(M - P) + 2M)/2$$

$$= 16.5M - 16.5P + M = 17.5M - 16.5P$$

$$10.5M + 5N - 17.5M + 16.5P = 135$$

$$45a + 66a - 7M = 135$$

$$111a - 7M = 135 \text{---(II)}$$

$$27M - 16P = 1300 \text{---(I)}$$

$$27M - 64a = 1300 \text{---(I)}$$

From equation (I) and (II)

$$I \times 7 \text{---} \rightarrow 189M - 448a = 9100$$

$$II \times 27 \text{---} \rightarrow 2997a - 189M = 3645$$

$$2549a = 12745$$

$$a = 5$$

and

$$M = 60$$

WKT

$$P = 4a \text{ and } N = 9a$$

$$N = 45$$

$$P = 20$$

Substitute all the variable values and get into the table.

@QuizByMickey

Cities	The total number of people going to each city	The number of people going by train	The number of people going by train
Delhi	1280	855	425
Patna	960	600	360
Ahmedabad	1100	620	480
Mumbai	1500	950	550
Goa	1320	720	600

Answer: D

Total number of people going to all five cities by flight = $425 + 360 + 480 + 550 + 600 = 2415$

= P% of 2415

= 20% of 2415

= 483

Average number of people going to all five cities together = $(1280 + 960 + 1100 + 1500 + 1320) / 5 = 1232$

= $(N + 5) / 2$ % of 1232

= 25% of 1232

= 308

Required Sum = $483 + 308 = 791$

Hence the answer is option D.

11. Questions

Number of people going to Delhi = $21M + P$

Number of people going to Goa = $33(M - P)$

Total number of people going to Delhi and Goa = 2600

$21M + P + 33M - 33P = 2600$

$54M - 32P = 2600$

$27M - 16P = 1300$ ---(I)

Total number of people going to Mumbai = $100N/3$

Number of people going to Mumbai by flight = $(100N/3 - 20P)/2$

Total number of people going to Ahmedabad = $20(N + 0.5P)$

Number of people going to Ahmedabad by flight = $(20(N + 0.5P) - 7P)/2$

@QuizByMickey

$$\frac{\left[\frac{100N/3 - 20P}{2}\right]}{\left[\frac{20(N + 0.5P) - 7P}{2}\right]} = \frac{55}{48}$$

$$(20N/3 - 4P) \times 48 = 11(20N + 10P - 7P)$$

$$320N - 192P = 220N + 33P$$

$$100N = 225P$$

$$P:N = 4:9$$

$$\text{Let } P = 4a \text{ and } N = 9a$$

$$\text{The number of people going to Delhi by train} = (21M + P + 10N - P)/2$$

$$= 10.5M + 5N$$

$$\text{The number of people going to Goa by train} = (33(M - P) + 2M)/2$$

$$= 16.5M - 16.5P + M = 17.5M - 16.5P$$

$$10.5M + 5N - 17.5M + 16.5P = 135$$

$$45a + 66a - 7M = 135$$

$$111a - 7M = 135 \text{---(II)}$$

$$27M - 16P = 1300 \text{---(I)}$$

$$27M - 64a = 1300 \text{---(I)}$$

From equation (I) and (II)

$$I \times 7 \text{ ---} \rightarrow 189M - 448a = 9100$$

$$II \times 27 \text{ ---} \rightarrow 2997a - 189M = 3645$$

$$2549a = 12745$$

$$a = 5$$

and

$$M = 60$$

WKT

$$P = 4a \text{ and } N = 9a$$

$$N = 45$$

$$P = 20$$

Substitute all the variable values and get into the table.

@QuizByMickey

Cities	The total number of people going to each city	The number of people going by train	The number of people going by train
Delhi	1280	855	425
Patna	960	600	360
Ahmedabad	1100	620	480
Mumbai	1500	950	550
Goa	1320	720	600

Answer: B

Total number of people going to Delhi = 1280

Number of males going to Delhi = 1280 - 542 = 738

Number of males going to Delhi by train = 855 * 60% = 513

Number of males going to Delhi by flight = 738 - 513 = 225

Require difference = 513 - 225 = 288

5M - 12 = 288

Hence the answer is option B.

12. Questions

Number of people going to Delhi = 21M + P

Number of people going to Goa = 33(M - P)

Total number of people going to Delhi and Goa = 2600

$21M + P + 33M - 33P = 2600$

$54M - 32P = 2600$

$27M - 16P = 1300$ ---(I)

Total number of people going to Mumbai = $100N/3$

Number of people going to Mumbai by flight = $(100N/3 - 20P)/2$ @QuizByMickey

Total number of people going to Ahmedabad = $20(N + 0.5P)$

Number of people going to Ahmedabad by flight = $(20(N + 0.5P) - 7P)/2$

$$\frac{\left[\frac{100N/3 - 20P}{2} \right]}{\left[\frac{20(N + 0.5P) - 7P}{2} \right]} = \frac{55}{48}$$

$(20N/3 - 4P) * 48 = 11(20N + 10P - 7P)$

$320N - 192P = 220N + 33P$

$$100N = 225P$$

$$P:N = 4:9$$

$$\text{Let } P = 4a \text{ and } N = 9a$$

$$\text{The number of people going to Delhi by train} = (21M + P + 10N - P)/2$$

$$= 10.5M + 5N$$

$$\text{The number of people going to Goa by train} = (33(M - P) + 2M)/2$$

$$= 16.5M - 16.5P + M = 17.5M - 16.5P$$

$$10.5M + 5N - 17.5M + 16.5P = 135$$

$$45a + 66a - 7M = 135$$

$$111a - 7M = 135 \text{---(II)}$$

$$27M - 16P = 1300 \text{---(I)}$$

$$27M - 64a = 1300 \text{---(I)}$$

From equation (I) and (II)

$$I * 7 \text{---} \rightarrow 189M - 448a = 9100$$

$$II * 27 \text{---} \rightarrow 2997a - 189M = 3645$$

$$2549a = 12745$$

$$a = 5$$

and

$$M = 60$$

WKT

$$P = 4a \text{ and } N = 9a$$

$$N = 45$$

$$P = 20$$

Substitute all the variable values and get into the table.

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Cities	The total number of people going to each city	The number of people going by train	The number of people going by train
Delhi	1280	855	425
Patna	960	600	360
Ahmedabad	1100	620	480
Mumbai	1500	950	550
Goa	1320	720	600

Answer: C

Average number of people going to Patna and Delhi = $(1280+960)/2 = 1120$

Average number of people going to Mumbai and Goa = $(1500+1320)/2 = 1410$

Required percentage

$$(1410 - 1120)/1410 \times 100$$

$$= 20.56\% \sim 21\%$$

Hence the answer is option C.

13. Questions

Answer: B

I). $x^3 + Ax^2 + 19x + 12 = 0$

Smallest prime number = 2

$$A = 8$$

$$x^3 + 8x^2 + 19x + 12 = 0$$

so,

coefficient of x^3 , x^2 , x and constant of the equation is 1, 8, 19 and 12 respectively.

$$\begin{array}{r|rrrr} -1 & 1 & 8 & 19 & 12 \\ & 0 & -1 & -7 & -12 \\ \hline & 1 & 7 & 12 & 0 \end{array}$$

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1

Let find the other roots

So, using (1, 7, and 12)

$$x^2 + 7x + 12 = 0$$

$$x^2 + 4x + 3x + 12 = 0$$

$$x(x + 4) + 3(x + 4) = 0$$

$$(x + 4)(x + 3) = 0$$

$$x = -1, -4 \text{ and } -3$$

II). $4y^3 - (B+12)y^2 + 68y - 2B = 0$

$$B = 8 \times 250\% = 20$$

$$4y^3 - 32y^2 + 68y - 40 = 0$$

so,

$$y^3 - 8y^2 + 17y - 10 = 0$$

coefficient of y^3 , y^2 , y and constant of the equation is 1, -8, 17 and -10 respectively.

1	1	-8	17	-10
	0	1	-7	10
	1	-7	10	0

Let find the other roots

So, using (1, -7, and 10)

$$y^2 - 7y + 10 = 0$$

$$y^2 - 5y - 2y + 10 = 0$$

$$y(y - 5) - 2(y - 5) = 0$$

$$(y - 5)(y - 2) = 0$$

$$y = 1, 5 \text{ and } 2$$

All roots of equation I are negative, and all roots of equation II are positive.

$$x < y$$

Hence the answer is option B.

14. Questions

Answer: E

(I): $3x^3 - 5x^2 - 11x = 3$

$$3x^3 - 5x^2 - 11x - 3 = 0$$

so,

coefficient of x^3 , x^2 , x and constant of the equation is 3, -5 -11 and -3 respectively.

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$$\begin{array}{r|rrrr} -1 & 3 & -5 & -11 & -3 \\ & 0 & -3 & 8 & 3 \\ \hline & 3 & -8 & -3 & 0 \end{array}$$

Let find the other roots

So, using (3, -8, and -3)

$$3x^2 - 8x - 3 = 0$$

$$3x^2 - 9x + x - 3 = 0$$

$$3x(x - 3) + (x - 3) = 0$$

$$(3x + 1)(x - 3) = 0$$

$$x = -1, 3 \text{ and } -1/3$$

(II): $2y^3 + y^2 - 5y = -2$

$$2y^3 + y^2 - 5y + 2 = 0$$

so,

coefficient of x^3 , x^2 , x and constant of the equation is 2, 1, -5 and 2 respectively.

$$\begin{array}{r|rrrr} 1 & 2 & 1 & -5 & 2 \\ & 0 & 2 & 3 & -2 \\ \hline & 2 & 3 & -2 & 0 \end{array}$$

Let find the other roots

So, using (2, 3, and -2)

$$2y^2 + 3y - 2 = 0$$

$$2y^2 + 4y - y - 2 = 0$$

$$2y(y + 2) - (y + 2) = 0$$

$$(2y - 1)(y + 2) = 0$$

$$y = 1, -2 \text{ and } 1/2$$

Relationship can't be determined

Hence the answer is option E.

15. Questions

Answer: E

(I): $x^3 + Qx^2 - (P-Q)x = 0$

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$$x(x^2 + Qx - (P - Q)) = 0$$

$$x = 0$$

and

$$x^2 + Qx - (P - Q) = 0$$

So,

$$-Q = -5$$

$$Q = 5$$

$$P = 3 \times 5 + 4 = 19$$

$$x^3 + 5x^2 - (19 - 5)x = 0$$

$$x^2 + 5x - 14 = 0$$

$$x^2 + 7x - 2x - 14 = 0$$

$$x(x+7) - 2(x+7) = 0$$

$$x = 2 \text{ or } -7$$

$$(II): y^3 - Ry^2 + (R-1)y = 0$$

$$y[y^2 - Ry + (R-1)] = 0$$

$$y = 0 \text{ and } y^2 - Ry + (R-1) = 0$$

$$R - 1 = 3$$

$$R = 4$$

$$y^2 - 4y + 3 = 0$$

$$y = 0, 3, \text{ and } 1$$

Relationship can't be determined

Hence the answer is option E.

16. Questions

Answer: E

$$I: X^2 + (P + 2)A - 12P = 0$$

In this equation belongs to one positive root and one negative root.

The product of the roots = $-12P$

The negative root = -12

So, the positive root = $(-12P/-12) = P$

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The sum of the roots = $-(P + 2)$

Now,

$$P + (-12) = -(P + 2)$$

$$P - 12 = -P - 2$$

$$2P = 14$$

$$P = 7$$

I: P is the prime number.

'P' is a prime number. This is satisfied.

II: The smallest root of the equation is a prime number.

The smallest root of the equation is -12 which is not a prime number.

III: 'P + 4' is a perfect square.

$$P + 4 = 5 + 4 = 9$$

9 is a perfect square. This is satisfied.

17. Questions

Answer: A

(I): $x^2 - 29x + 52A = 0$

One of the root is 16

$$16^2 - 29 * 16 + 52A = 0$$

$$256 - 464 + 52A = 0$$

$$52A = 208$$

$$A = 4$$

Let

$$x^2 - 29x + 52 * 4 = 0$$

$$x^2 - 29x + 208 = 0$$

$$x^2 - 13x - 16x + 208 = 0$$

$$x(x - 13) - 16(x - 13) = 0$$

$$(x - 13)(x - 16) = 0$$

$$x = 13 \text{ and } 16$$

The smallest root of the equation I is 13

And also the smallest root of the equation II is 13

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$$(II): y^2 - 38y + 65B = 0$$

$$13^2 - 38 * 13 + 65B = 0$$

$$169 - 494 + 65B = 0$$

$$65B = 325$$

$$B = 5$$

Let

$$y^2 - 38y + 65 * 5 = 0$$

$$y^2 - 38y + 325 = 0$$

$$y^2 - 13y - 25y + 325 = 0$$

$$y(y - 13) - 25(y - 13) = 0$$

$$(y - 13)(y - 25) = 0$$

$$y = 13 \text{ and } 25$$

Thus, sum of the positive roots of the both equations = $16 + 25 = 41$

18. Questions

Length of train A = 150 m

Length of train B = $(156 - X/2)m$

Speed of train A = $(150+X)/9$ mps

Speed of train B = $(Y - 20)$ km/h

Time taken by train B to cross a pole = 5.76 seconds

$$(Y - 20) * 5/18 * 5.76 = 156 - X/2$$

$$2(1.6Y - 32) = 312 - X$$

$$3.2Y - 64 = 312 - X$$

$$3.2Y + X = 376 \text{ ---(I)}$$

$$\text{Length of train B} = (Y - 20) * 5/18 * 5.76 = 1.6Y - 32$$

Let, Time taken by train B to cover the length of the platform = 14.4 Seconds

$$(Y - 20) * 5/18 * 14.4 = 1.6Y - 32 + X + 24$$

$$4Y - 80 = 1.6Y - 32 + X + 24$$

$$2.4Y - X = 72 \text{ ---(II)}$$

From equation I and II

$$Y = 80 \text{ and } X = 120$$

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Length of train B = 96 m

Speed of train A = 30 mps

Speed of train B = 60 kmph

Speed of train C = 20 mps

Let the length of train C = C metre

Train B and train C both running in same directions can cross each other in 82.8 seconds

$$(20 - 60 \times 5/18) \times 82.8 = 96 + C$$

$$276 - 96 = C$$

$$C = 180$$

Length of train C = 180 metre

Answer: D

Train C reach station Q at 5:20 P.m. If it did not stop it would have reached at 4:20 p.m.

Time taken by train C to reach station Q without any stop = 8 hours 20 minutes

$$(20 \times 18/5 \times 25/3) = \text{distance between station P and Q}$$

$$600 \text{ km} = \text{distance between station P and Q}$$

$$\text{Time taken by train A to cover 600 km distance} = 600/108 = 50/9 \text{ hours}$$

$$\text{Time taken by train B to cover 600 km distance} = 600/60 = 10 \text{ hours}$$

$$\text{Required difference} = (10 - 50/9) \text{ hours}$$

$$40/9 \text{ hours}$$

$$= 4 \frac{4}{9} \text{ hours}$$

Hence the answer is option D.

19. Questions

Length of train A = 150 m

Length of train B = $(156 - X/2)$ m

Speed of train A = $(150 + X)/9$ mps

Speed of train B = $(Y - 20)$ km/h

Time taken by train B to cross a pole = 5.76 seconds

$$(Y - 20) \times 5/18 \times 5.76 = 156 - X/2$$

$$2(1.6Y - 32) = 312 - X$$

$$3.2Y - 64 = 312 - X$$

$$3.2Y + X = 376 \text{ ---(I)}$$

@QuizByMickey

$$\text{Length of train B} = (Y - 20) * 5/18 * 5.76 = 1.6Y - 32$$

Let, Time taken by train B to cover the length of the platform = 14.4 Seconds

$$(Y - 20) * 5/18 * 14.4 = 1.6Y - 32 + X + 24$$

$$4Y - 80 = 1.6Y - 32 + X + 24$$

$$2.4Y - X = 72 \text{---(II)}$$

From equation I and II

$$Y = 80 \text{ and } X = 120$$

Length of train B = 96 m

Speed of train A = 30 mps

Speed of train B = 60 kmph

Speed of train C = 20 mps

Let the length of train C = C metre

Train B and train C both running in same directions can cross each other in 82.8 seconds

$$(20 - 60 * 5/18) * 82.8 = 96 + C$$

$$276 - 96 = C$$

$$C = 180$$

Length of train C = 180 metre

Answer: B

Speed of train B = 60 km/h

Length of train B = 96 m

Let the time taken by train B to completely cross the person = t seconds

$$(60 * 5/18 - 6) * t = 96$$

$$32t = 288$$

$$t = 9 \text{ seconds}$$

Hence the answer is option B.

20. Questions

Length of train A = 150 m

Length of train B = $(156 - X/2)$ m

Speed of train A = $(150 + X)/9$ mps

Speed of train B = $(Y - 20)$ km/h

Time taken by train B to cross a pole = 5.76 seconds

@QuizByMickey

$$(Y - 20) * 5/18 * 5.76 = 156 - X/2$$

$$2(1.6Y - 32) = 312 - X$$

$$3.2Y - 64 = 312 - X$$

$$3.2Y + X = 376 \text{ ---(I)}$$

$$\text{Length of train B} = (Y - 20) * 5/18 * 5.76 = 1.6Y - 32$$

Let, Time taken by train B to cover the length of the platform = 14.4 Seconds

$$(Y - 20) * 5/18 * 14.4 = 1.6Y - 32 + X + 24$$

$$4Y - 80 = 1.6Y - 32 + X + 24$$

$$2.4Y - X = 72 \text{ ---(II)}$$

From equation I and II

$$Y = 80 \text{ and } X = 120$$

$$\text{Length of train B} = 96 \text{ m}$$

$$\text{Speed of train A} = 30 \text{ mps}$$

$$\text{Speed of train B} = 60 \text{ kmph}$$

$$\text{Speed of train C} = 20 \text{ mps}$$

Let the length of train C = C metre

Train B and train C both running in same directions can cross each other in 82.8 seconds

$$(20 - 60 * 5/18) * 82.8 = 96 + C$$

$$276 - 96 = C$$

$$C = 180$$

$$\text{Length of train C} = 180 \text{ metre}$$

Answer: C

$$\text{Length of bridge} = X = 120 \text{ m}$$

$$\text{Length of platform} = X + 24 = 144 \text{ m}$$

$$\text{Time taken by train C to cross the bridge} = (120 + 180)/20 = 15 \text{ seconds}$$

$$M = 15$$

$$\text{Time taken by train A to cross the platform} = (144 + 150)/30 = 9.8 \text{ seconds}$$

$$N + 4.8 = 9.8$$

$$N = 5$$

$$\text{The time taken by train C to cross a tunnel} = (180 + 150)/20$$

$$= 16.5 \text{ seconds}$$

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= $M/10+3N$ seconds

Hence the answer is option C

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